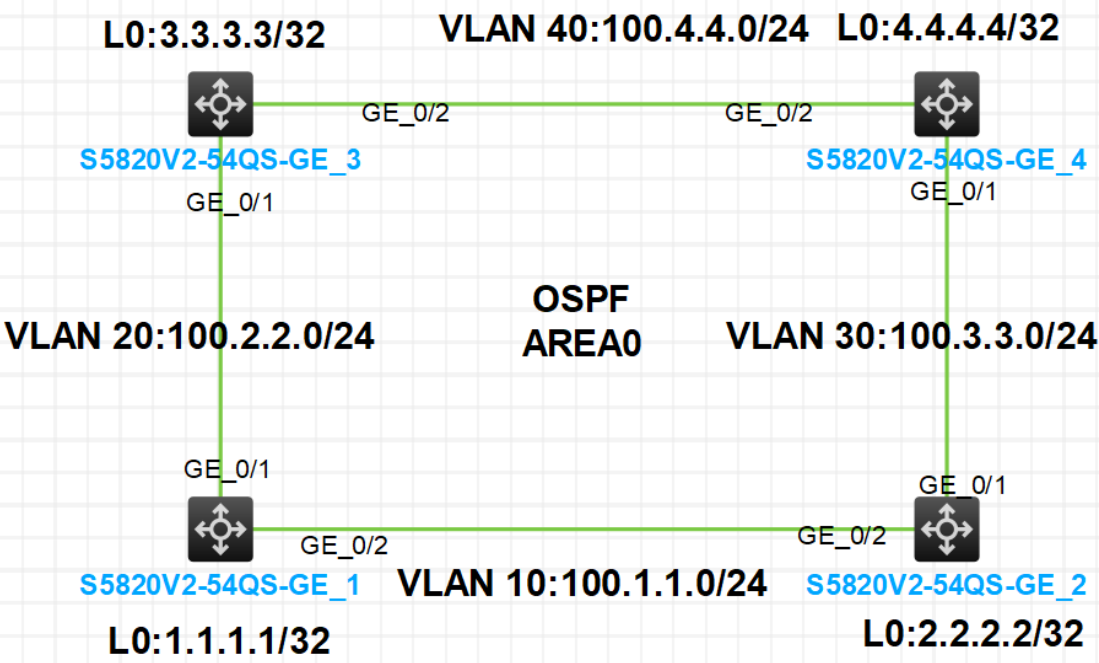


VLAN路由实验

实验拓扑

- 实验需求
- 1.按照图示创建 Vlan，所有接口配置为 Access 类型加入相应互联 VLAN
 - 2.按照图示配置 IP 地址
 - 3.配置 OSPF，实现全网互通



实验需求

- 1.按照图示创建 Vlan，所有接口配置为 Access 类型加入相应互联 VLAN
- 2.按照图示配置 IP 地址
- 3.配置 OSPF，实现全网互通

实验配置

- ```
1 1.配置IP地址
2 [SW1]vlan 10
3 [SW1-vlan10]vlan 20
4 [SW1-vlan20]int vlan 10
5 [SW1-Vlan-interface10]ip add 100.1.1.1 24
6 [SW1-Vlan-interface10]int vlan 20
```

```
7 [SW1-Vlan-interface20]ip add 100.2.2.1 24
8 [SW1-Vlan-interface20]int l0
9 [SW1-LoopBack0]ip ad 1.1.1.1 32
10 [SW1-LoopBack0]int g1/0/1
11 [SW1-GigabitEthernet1/0/1]po link-type access
12 [SW1-GigabitEthernet1/0/1]po access vlan 20
13 [SW1-GigabitEthernet1/0/1]int g1/0/2
14 [SW1-GigabitEthernet1/0/2]po link-type access
15 [SW1-GigabitEthernet1/0/2]po access vlan 10
16 ----
17 [SW2]vlan 10
18 [SW2-vlan10]vlan 30
19 [SW2-vlan20]int vlan 10
20 [SW2-Vlan-interface10]ip add 100.1.1.2 24
21 [SW2-Vlan-interface10]int vlan 30
22 [SW2-Vlan-interface20]ip add 100.3.3.2 24
23 [SW2-Vlan-interface20]int l0
24 [SW2-LoopBack0]ip ad 2.2.2.2 32
25 [SW2-LoopBack0]int g1/0/1
26 [SW2-GigabitEthernet1/0/1]po link-type access
27 [SW2-GigabitEthernet1/0/1]po access vlan 30
28 [SW2-GigabitEthernet1/0/1]int g1/0/2
29 [SW2-GigabitEthernet1/0/2]po link-type access
30 [SW2-GigabitEthernet1/0/2]po access vlan 10
31 ---
32 [SW3]vlan 20
33 [SW3-vlan20]vlan 40
34 [SW3-vlan40]int vlan 20
35 [SW3-Vlan-interface20]ip add 100.2.2.3 24
36 [SW3-Vlan-interface20]int vlan 40
37 [SW3-Vlan-interface40]ip add 100.4.4.3 24
38 [SW3-Vlan-interface40]int l0
39 [SW3-LoopBack0]ip ad 3.3.3.3 32
40 [SW3-LoopBack0]int g1/0/1
41 [SW3-GigabitEthernet1/0/1]po link-type access
42 [SW3-GigabitEthernet1/0/1]po access vlan 20
43 [SW3-GigabitEthernet1/0/1]int g1/0/2
44 [SW3-GigabitEthernet1/0/2]po link-type access
45 [SW3-GigabitEthernet1/0/2]po access vlan 40
46 ---
47 [SW4]vlan 30
48 [SW4-vlan30]vlan 40
49 [SW4-vlan40]int vlan 30
50 [SW4-Vlan-interface30]ip add 100.3.3.4 24
51 [SW4-Vlan-interface30]int vlan 40
52 [SW4-Vlan-interface40]ip add 100.4.4.4 24
53 [SW4-Vlan-interface40]int l0
54 [SW4-LoopBack0]ip ad 4.4.4.4 32
55 [SW4-LoopBack0]int g1/0/1
56 [SW4-GigabitEthernet1/0/1]po link-type access
57 [SW4-GigabitEthernet1/0/1]po access vlan 30
58 [SW4-GigabitEthernet1/0/1]int g1/0/2
59 [SW4-GigabitEthernet1/0/2]po link-type access
```

```
60 [SW4-GigabitEthernet1/0/2]po access vlan 40
61
62 [SW1-LoopBack0]ospf 1 rou 1.1.1.1
63 [SW1-ospf-1]a 0
64 [SW1-ospf-1-area-0.0.0.0]net 1.1.1.1 0.0.0.0
65 [SW1-ospf-1-area-0.0.0.0]net 100.1.1.0 0.0.0.255
66 [SW1-ospf-1-area-0.0.0.0]net 100.2.2.0 0.0.0.255
67 ---
68 [SW2-GigabitEthernet1/0/2]ospf 1 rou 2.2.2.2
69 [SW2-ospf-1]a 0
70 [SW2-ospf-1-area-0.0.0.0]net 2.2.2.2 0.0.0.0
71 [SW2-ospf-1-area-0.0.0.0]net 100.1.1.0 0.0.0.255
72 [SW2-ospf-1-area-0.0.0.0]net 100.3.3.0 0.0.0.255
73 ---
74 [SW3-GigabitEthernet1/0/2]ospf 1 rou 3.3.3.3
75 [SW3-ospf-1]a 0
76 [SW3-ospf-1-area-0.0.0.0]net 3.3.3.3 0.0.0.0
77 [SW3-ospf-1-area-0.0.0.0]net 100.2.2.0 0.0.0.255
78 [SW3-ospf-1-area-0.0.0.0]net 100.4.4.0 0.0.0.255
79 ---
80 [SW4-GigabitEthernet1/0/2]ospf 1 rou 4.4.4.4
81 [SW4-ospf-1]a 0
82 [SW4-ospf-1-area-0.0.0.0]net 4.4.4.4 0.0.0.0
83 [SW4-ospf-1-area-0.0.0.0]net 100.3.3.0 0.0.0.255
84 [SW4-ospf-1-area-0.0.0.0]net 100.4.4.0 0.0.0.255
85
```

## 结果验证

### 1.连通性测试

```
1 <SW1>ping 4.4.4.4
2 Ping 4.4.4.4 (4.4.4.4): 56 data bytes, press CTRL_C to break
3 56 bytes from 4.4.4.4: icmp_seq=0 ttl=254 time=0.895 ms
4 56 bytes from 4.4.4.4: icmp_seq=1 ttl=254 time=1.378 ms
5 ---
6 <SW2>ping 3.3.3.3
7 Ping 3.3.3.3 (3.3.3.3): 56 data bytes, press CTRL_C to break
8 56 bytes from 3.3.3.3: icmp_seq=0 ttl=254 time=1.081 ms
9 56 bytes from 3.3.3.3: icmp_seq=1 ttl=254 time=1.147 ms
10 56 bytes from 3.3.3.3: icmp_seq=2 ttl=254 time=0.944 ms
```

### 2.查看IP地址

```
1 <SW1>display ip interface brief
2 *down: administratively down
```

|    |                                 |          |          |            |             |
|----|---------------------------------|----------|----------|------------|-------------|
| 3  | (s): spoofing (l): loopback     |          |          |            |             |
| 4  | Interface                       | Physical | Protocol | IP Address | Description |
| 5  | Loop0                           | up       | up(s)    | 1.1.1.1    | --          |
| 6  | MGE0/0/0                        | down     | down     | --         | --          |
| 7  | Vlan10                          | up       | up       | 100.1.1.1  | --          |
| 8  | Vlan20                          | up       | up       | 100.2.2.1  | --          |
| 9  | ---                             |          |          |            |             |
| 10 | display ip interface brief      |          |          |            |             |
| 11 | *down: administratively down    |          |          |            |             |
| 12 | (s): spoofing (l): loopback     |          |          |            |             |
| 13 | Interface                       | Physical | Protocol | IP Address | Description |
| 14 | Loop0                           | up       | up(s)    | 2.2.2.2    | --          |
| 15 | MGE0/0/0                        | down     | down     | --         | --          |
| 16 | Vlan10                          | up       | up       | 100.1.1.2  | --          |
| 17 | Vlan30                          | up       | up       | 100.3.3.2  | --          |
| 18 | ---                             |          |          |            |             |
| 19 | display ip interface brief      |          |          |            |             |
| 20 | *down: administratively down    |          |          |            |             |
| 21 | (s): spoofing (l): loopback     |          |          |            |             |
| 22 | Interface                       | Physical | Protocol | IP Address | Description |
| 23 | Loop0                           | up       | up(s)    | 3.3.3.3    | --          |
| 24 | MGE0/0/0                        | down     | down     | --         | --          |
| 25 | Vlan20                          | up       | up       | 100.2.2.3  | --          |
| 26 | Vlan40                          | up       | up       | 100.4.4.3  | --          |
| 27 | ---                             |          |          |            |             |
| 28 | <SW4>display ip interface brief |          |          |            |             |
| 29 | *down: administratively down    |          |          |            |             |
| 30 | (s): spoofing (l): loopback     |          |          |            |             |
| 31 | Interface                       | Physical | Protocol | IP Address | Description |
| 32 | Loop0                           | up       | up(s)    | 4.4.4.4    | --          |
| 33 | MGE0/0/0                        | down     | down     | --         | --          |
| 34 | Vlan30                          | up       | up       | 100.3.3.4  | --          |
| 35 | Vlan40                          | up       | up       | 100.4.4.4  | --          |

### 3.查看VLAN

```

1 <SW1>display vlan brief
2 Brief information about all VLANs:
3 Supported Minimum VLAN ID: 1
4 Supported Maximum VLAN ID: 4094
5 Default VLAN ID: 1
6
7 VLAN ID Name
8
9 1 VLAN 0001
10
11
12
13
14

```

```

15 GE1/0/26 GE1/0/27 GE1/0/28
16 GE1/0/29 GE1/0/30 GE1/0/31
17 GE1/0/32 GE1/0/33 GE1/0/34
18 GE1/0/35 GE1/0/36 GE1/0/37
19 GE1/0/38 GE1/0/39 GE1/0/40
20 GE1/0/41 GE1/0/42 GE1/0/43
21 GE1/0/44 GE1/0/45 GE1/0/46
22 GE1/0/47 GE1/0/48 XGE1/0/49
23 XGE1/0/50 XGE1/0/51 XGE1/0/52
24 10 VLAN 0010 GE1/0/2
25 20 VLAN 0020 GE1/0/1
26 ---
27 <SW2>display vlan brief
28 Brief information about all VLANs:
29 Supported Minimum VLAN ID: 1
30 Supported Maximum VLAN ID: 4094
31 Default VLAN ID: 1
32 VLAN ID Name Port
33 1 VLAN 0001 FGE1/0/53 FGE1/0/54 GE1/0/3
34 GE1/0/4 GE1/0/5 GE1/0/6 GE1/0/7
35 GE1/0/8 GE1/0/9 GE1/0/10
36 GE1/0/11 GE1/0/12 GE1/0/13
37 GE1/0/14 GE1/0/15 GE1/0/16
38 GE1/0/17 GE1/0/18 GE1/0/19
39 GE1/0/20 GE1/0/21 GE1/0/22
40 GE1/0/23 GE1/0/24 GE1/0/25
41 GE1/0/26 GE1/0/27 GE1/0/28
42 GE1/0/29 GE1/0/30 GE1/0/31
43 GE1/0/32 GE1/0/33 GE1/0/34
44 GE1/0/35 GE1/0/36 GE1/0/37
45 GE1/0/38 GE1/0/39 GE1/0/40
46 GE1/0/41 GE1/0/42 GE1/0/43
47 GE1/0/44 GE1/0/45 GE1/0/46
48 GE1/0/47 GE1/0/48 XGE1/0/49
49 XGE1/0/50 XGE1/0/51 XGE1/0/52
50 10 VLAN 0010 GE1/0/2
51 30 VLAN 0030 GE1/0/1
52 ---
53 <SW3>display vlan brief
54 Brief information about all VLANs:
55 Supported Minimum VLAN ID: 1
56 Supported Maximum VLAN ID: 4094
57 Default VLAN ID: 1
58 VLAN ID Name Port
59 1 VLAN 0001 FGE1/0/53 FGE1/0/54 GE1/0/3
60 GE1/0/4 GE1/0/5 GE1/0/6 GE1/0/7
61 GE1/0/8 GE1/0/9 GE1/0/10
62 GE1/0/11 GE1/0/12 GE1/0/13
63 GE1/0/14 GE1/0/15 GE1/0/16
64 GE1/0/17 GE1/0/18 GE1/0/19
65 GE1/0/20 GE1/0/21 GE1/0/22
66 GE1/0/23 GE1/0/24 GE1/0/25
67 GE1/0/26 GE1/0/27 GE1/0/28

```

|     |                                    |           |           |           |           |
|-----|------------------------------------|-----------|-----------|-----------|-----------|
| 68  |                                    |           | GE1/0/29  | GE1/0/30  | GE1/0/31  |
| 69  |                                    |           | GE1/0/32  | GE1/0/33  | GE1/0/34  |
| 70  |                                    |           | GE1/0/35  | GE1/0/36  | GE1/0/37  |
| 71  |                                    |           | GE1/0/38  | GE1/0/39  | GE1/0/40  |
| 72  |                                    |           | GE1/0/41  | GE1/0/42  | GE1/0/43  |
| 73  |                                    |           | GE1/0/44  | GE1/0/45  | GE1/0/46  |
| 74  |                                    |           | GE1/0/47  | GE1/0/48  | XGE1/0/49 |
| 75  |                                    |           | XGE1/0/50 | XGE1/0/51 | XGE1/0/52 |
| 76  | 20                                 | VLAN 0020 | GE1/0/1   |           |           |
| 77  | 40                                 | VLAN 0040 | GE1/0/2   |           |           |
| 78  | ---                                |           |           |           |           |
| 79  | <SW4>display vlan brief            |           |           |           |           |
| 80  | Brief information about all VLANs: |           |           |           |           |
| 81  | Supported Minimum VLAN ID: 1       |           |           |           |           |
| 82  | Supported Maximum VLAN ID: 4094    |           |           |           |           |
| 83  | Default VLAN ID: 1                 |           |           |           |           |
| 84  | VLAN ID                            | Name      | Port      |           |           |
| 85  | 1                                  | VLAN 0001 | FGE1/0/53 | FGE1/0/54 | GE1/0/3   |
| 86  |                                    |           | GE1/0/4   | GE1/0/5   | GE1/0/6   |
| 87  |                                    |           | GE1/0/8   | GE1/0/9   | GE1/0/10  |
| 88  |                                    |           | GE1/0/11  | GE1/0/12  | GE1/0/13  |
| 89  |                                    |           | GE1/0/14  | GE1/0/15  | GE1/0/16  |
| 90  |                                    |           | GE1/0/17  | GE1/0/18  | GE1/0/19  |
| 91  |                                    |           | GE1/0/20  | GE1/0/21  | GE1/0/22  |
| 92  |                                    |           | GE1/0/23  | GE1/0/24  | GE1/0/25  |
| 93  |                                    |           | GE1/0/26  | GE1/0/27  | GE1/0/28  |
| 94  |                                    |           | GE1/0/29  | GE1/0/30  | GE1/0/31  |
| 95  |                                    |           | GE1/0/32  | GE1/0/33  | GE1/0/34  |
| 96  |                                    |           | GE1/0/35  | GE1/0/36  | GE1/0/37  |
| 97  |                                    |           | GE1/0/38  | GE1/0/39  | GE1/0/40  |
| 98  |                                    |           | GE1/0/41  | GE1/0/42  | GE1/0/43  |
| 99  |                                    |           | GE1/0/44  | GE1/0/45  | GE1/0/46  |
| 100 |                                    |           | GE1/0/47  | GE1/0/48  | XGE1/0/49 |
| 101 |                                    |           | XGE1/0/50 | XGE1/0/51 | XGE1/0/52 |
| 102 | 30                                 | VLAN 0030 | GE1/0/1   |           |           |
| 103 | 40                                 | VLAN 0040 | GE1/0/2   |           |           |

### 3.OSPF邻居表

|    |                                       |           |     |           |         |           |
|----|---------------------------------------|-----------|-----|-----------|---------|-----------|
| 1  | <SW1>display ospf peer                |           |     |           |         |           |
| 2  |                                       |           |     |           |         |           |
| 3  | OSPF Process 1 with Router ID 1.1.1.1 |           |     |           |         |           |
| 4  | Neighbor Brief Information            |           |     |           |         |           |
| 5  |                                       |           |     |           |         |           |
| 6  | Area: 0.0.0.0                         |           |     |           |         |           |
| 7  | Router ID                             | Address   | Pri | Dead-Time | State   | Interface |
| 8  | 2.2.2.2                               | 100.1.1.2 | 1   | 30        | Full/DR | Vlan10    |
| 9  | 3.3.3.3                               | 100.2.2.3 | 1   | 39        | Full/DR | Vlan20    |
| 10 | ---                                   |           |     |           |         |           |
| 11 | <SW2>display ospf peer                |           |     |           |         |           |

```
12
13 OSPF Process 1 with Router ID 2.2.2.2
14 Neighbor Brief Information
15
16 Area: 0.0.0.0
17 Router ID Address Pri Dead-Time State Interface
18 1.1.1.1 100.1.1.1 1 34 Full/BDR Vlan10
19 4.4.4.4 100.3.3.4 1 38 Full/DR Vlan30
20 ---
21 <SW3>display ospf peer
22
23 OSPF Process 1 with Router ID 3.3.3.3
24 Neighbor Brief Information
25
26 Area: 0.0.0.0
27 Router ID Address Pri Dead-Time State Interface
28 1.1.1.1 100.2.2.1 1 31 Full/BDR Vlan20
29 4.4.4.4 100.4.4.4 1 37 Full/DR Vlan40
30 ---
31 <SW4>display ospf peer
32
33 OSPF Process 1 with Router ID 4.4.4.4
34 Neighbor Brief Information
35
36 Area: 0.0.0.0
37 Router ID Address Pri Dead-Time State Interface
38 2.2.2.2 100.3.3.2 1 34 Full/BDR Vlan30
39 3.3.3.3 100.4.4.3 1 36 Full/BDR Vlan40
```

---

# 附录

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